

CRANBROOK SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2001

MATHEMATICS

4 UNIT (Additional)

Time allowed – Three hours

DIRECTIONS TO CANDIDATES

- * Attempt all questions.
 - * ALL questions are of equal value.
 - * All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
 - * Standard integrals are printed on the back page.
 - * Board-approved calculators may be used.
 - * You may ask for extra Writing Booklets if you need them.
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- * Submit your work in four 8 page booklets :
 - (i) QUESTIONS 1 & 2
 - (ii) QUESTIONS 3 & 4
 - (iii) QUESTIONS 5 & 6
 - (iv) QUESTIONS 7 & 8

1. (8 page booklet)

- (a) Find (i) $\int \cot x \operatorname{cosec}^2 x \, dx$ (ii) $\int \frac{\sec^2 x}{3 - \tan x} dx$ [4 marks]
- (b) Prove that $\int_{\frac{\pi}{2}}^{\frac{5}{2}} \frac{dx}{\sqrt{(x-5)(7-x)}} = \frac{\pi}{3}$, by using the substitution $u = x - 6$. [3 marks]
- (c) (i) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. [2 marks]
- (ii) Hence or otherwise evaluate $\int_0^{\frac{\pi}{2}} \frac{\cos^5 x}{\cos^5 x + \sin^5 x} dx$. [2 marks]
- (d) Evaluate $\int_0^{\frac{\pi}{4}} \frac{dx}{2 \sin 2x + \cos x}$ [4 marks]

2.

- (a) Evaluate $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x^3}{\cos x} dx$ [2 marks]
- (b) Find $\int \sin^3 2x \cos^2 2x \, dx$ [3 marks]
- (c) Find $\int \frac{4x-3}{\sqrt{6+2x-3x^2}} dx$ [4 marks]
- (d) If $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \sin^2 x \, dx$ for $n \geq 0$, show that $I_n = \frac{n-1}{n+2} I_{n-2}$ for $n \geq 2$. [4 marks]
- Hence or otherwise evaluate $\int_0^{\frac{\pi}{2}} \cos^4 x \sin^2 x \, dx$. [2 marks]

3. (new 8 page booklet please)

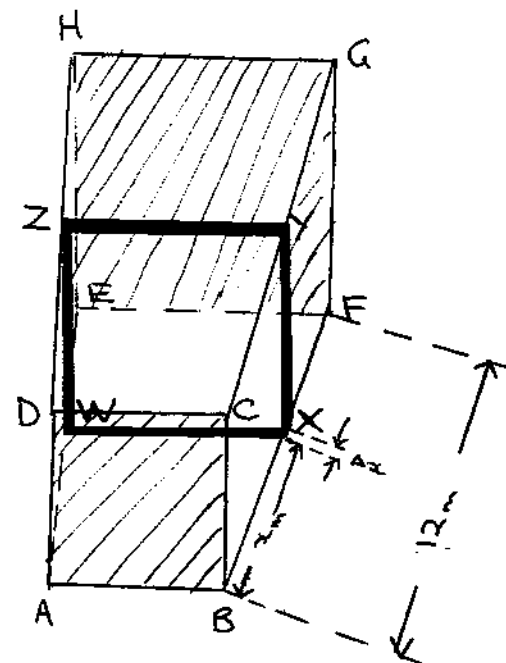
- (a) (i) Given $z_1 = 1-i$ and $z_2 = -1+\sqrt{3}i$ evaluate $|z_1 z_2|$ and $\arg(z_1 z_2)$
- (ii) Find $z_1 z_2$ in cartesian form, and hence show that $\cos \frac{5\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$ [6 marks]
- (b) If z is a complex number for which $|z|=1$ show that
- (i) $1 \leq |z+2| \leq 3$ and (ii) $-\frac{\pi}{6} \leq \arg(z+2) \leq \frac{\pi}{6}$ [4 marks]
- (c) (i) Given that $z + \frac{1}{z} = k$, a real number, show that z lies either on the real axis or on the unit circle, centre the origin.
- (ii) If z lies on the real axis, show that $|k| \geq 2$; if z lies on the unit circle, show that $|k| \leq 2$. [5 marks]

4.

- (a) Find integers a and b such that $(x+1)^2$ is a factor of $x^5 + 2x^2 + ax + b$. [3 marks]
- (b) The equation $z^2 + (1+i)z + k = 0$ has a root $1-2i$. Find the other root, and the value of k . [3 marks]
- (c) Let α, β, γ be the roots (none of which is zero) of $x^3 + 3px + q = 0$.
- (i) Obtain the monic equation whose roots are $\frac{\alpha\beta}{\gamma}, \frac{\beta\gamma}{\alpha}, \frac{\gamma\alpha}{\beta}$.
- (ii) Deduce that $\gamma = \alpha\beta$ if and only if $(3p-q)^2 + q = 0$ [9 marks]

5. (new 8 page booklet please)

- (a) The region bounded by the circle $x^2 + y^2 = 4$ and the parabola $y^2 = 3x$ is rotated about the x -axis. By including appropriate diagrams in each case, find the volume of the solid of revolution by using:
- (i) circular discs [5 marks]
- (ii) cylindrical shells. [5 marks]
- (b) In the solid shown ABCD and EFGH are squares of side 6 m and 10 m respectively. BCGF is a ~~parallelogram~~ of height 12 m. Cross-sections parallel to the ends are squares. Show that at a distance x m from the base AB the area of the cross-section is $\left(6 + \frac{x}{3}\right)^2$. Hence, by taking slices of thickness Δx find the total volume of the solid. [5 marks]



6.

- (a) $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ are points on the rectangular hyperbola $xy = 9$. The equation of chord PQ is $x + pqy = 3(p + q)$.

(i) Find the co-ordinates of N , the midpoint of PQ .

(ii) If chord PQ is a tangent to the parabola $y^2 = 3x$ prove that the locus of N is $3x = -8y^2$.

[5 marks]

- (b) A cylinder of constant volume V has its radius increasing at 5% per minute. At what % rate is the height diminishing?

[4 marks]

- (c) A cyclist and a jogger journey along two roads OA and OB , which are inclined at 60° to one another. The cyclist starts at a point P , 10 km from O along OA and cycles towards O . At the same instant the jogger starts from O and runs away from O along OB . If the cyclist travels at 8 km/h and the jogger runs at 5 km/h find the rate at which the distance between the two is changing after 90 minutes (in km/h), correct to 2 decimal places.

[6 marks]

7. (new 8 page booklet please)

- (a) Show that the ellipse $4x^2 + 9y^2 = 36$ and the hyperbola $4x^2 - y^2 = 4$ intersect at right angles. [5 marks]

- (b) You are given that the equation of the normal to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ ($a^2 > b^2$) at the point $P(x_1, y_1)$ is $a^2y_1x - b^2x_1y = (a^2 - b^2)x_1y_1$.

- (i) This normal meets the major axis of the ellipse at G . S is a focus of the ellipse. Show that $GS = e \times PS$, where e is the eccentricity of the ellipse.

[5 marks]

- (ii) The normal at the point $P(5\cos\theta, 3\sin\theta)$ on $\frac{x^2}{25} + \frac{y^2}{9} = 1$ cuts the major and minor axes of the ellipse at G and H respectively. Show that as P moves on the ellipse, the mid-point of GH describes another ellipse with the same eccentricity as the first.

[5 marks]

8.

- (a) In a certain cricket club there are 15 players available for selection, including 2 Smith brothers, 3 Brown brothers and 10 others. In how many ways may an eleven be selected for a game, if no more than 1 Smith and 2 Browns may be chosen?

[3 marks]

- (b) Given that $\sin^{-1}x$, $\cos^{-1}x$ and $\sin^{-1}(1-x)$ are all acute

(i) show that $\sin[\sin^{-1}x - \cos^{-1}x] = 2x^2 - 1$

(ii) solve the equation $\sin^{-1}x - \cos^{-1}x = \sin^{-1}(1-x)$

[5 marks]

- (c) The equation of a curve is $x^2y^2 - x^2 + y^2 = 0$.

(i) Show that the numerical value of y is always less than 1.

(ii) Find the equations of the asymptotes.

(iii) Show that $\frac{dy}{dx} = \frac{y^3}{x^3}$

(iv) Sketch the curve.

[7 marks]

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} \quad (n \neq -1; x \neq 0 \text{ if } n < 0)$$

$$\int \frac{1}{x} dx = \log_e x \quad (x > 0)$$

$$\int \cos ax dx = \frac{1}{a} \sin ax \quad (a \neq 0)$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax \quad (a \neq 0)$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} \quad (a \neq 0)$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} \quad (a > 0, -a < x < a)$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log_e \left\{ x + \sqrt{x^2 - a^2} \right\} \quad (|x| > |a|)$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log_e \left\{ x + \sqrt{x^2 + a^2} \right\}$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} \quad (a \neq 0)$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax \quad (a \neq 0)$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax \quad (a \neq 0)$$

4 UNIT: CRAWFORD TRIAL 2001 SOLUTIONS

(a) (i) $I = \int \cot x \operatorname{cosec}^2 x \, dx$
 let $u = \cot x$
 $\therefore \frac{du}{dx} = -\operatorname{cosec}^2 x$
 $\therefore I = \int u \cdot -du$
 $= -\frac{u^2}{2} + C$
 $= -\frac{\cot^2 x}{2} + C$

(ii) $I = \int \frac{\sec^2 x}{3 - \tan x} \, dx$

let $u = 3 - \tan x$
 $\frac{du}{dx} = -\sec^2 x$

$\therefore I = \int \frac{-du}{u}$
 $= -\ln|u| + C$
 $= -\ln|3 - \tan x| + C$

(b) $I = \int_{5\frac{1}{2}}^{6\frac{1}{2}} \frac{dx}{\sqrt{(x-5)(7-x)}}$

let $u = x - 6$ when $x = 5\frac{1}{2}$ $u = -\frac{1}{2}$
 $\therefore \frac{du}{dx} = 1$ $x = 6\frac{1}{2}$ $u = \frac{1}{2}$

$\therefore I = \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{du}{\sqrt{(u+1)(1-u)}}$
 $= \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{du}{\sqrt{1-u^2}}$
 $= 2 \int_0^{\frac{1}{2}} \frac{du}{\sqrt{1-u^2}}$
 $= 2 [\sin^{-1} u]_0^{\frac{1}{2}}$
 $= 2 \left[\frac{\pi}{6} - 0 \right] = \frac{\pi}{3}$

(c) (i) TO PROVE: $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$
 PROOF: LHS = $\int_0^a f(x) \, dx$
 let $x = a - u$ when $x = a$ $u = 0$
 $\therefore \frac{dx}{du} = -1$ $x = 0$ $u = a$

$\therefore \text{LHS} = \int_a^0 f(a-u) \cdot -du$
 $= \int_0^a f(a-u) \cdot du$
 $= \int_0^a f(a-x) \, dx$
 (reverting to the variable x)
 $= \text{RHS}$

(ii) $I = \int_0^{\frac{\pi}{2}} \frac{\cos^5 x \, dx}{\cos^5 x + \sin^5 x}$
 $= \int_0^{\frac{\pi}{2}} \frac{\cos^5(\frac{\pi}{2} - x) \, dx}{\cos^5(\frac{\pi}{2} - x) + \sin^5(\frac{\pi}{2} - x)}$
 $= \int_0^{\frac{\pi}{2}} \frac{\sin^5 x \, dx}{\sin^5 x + \cos^5 x}$

$\therefore 2I = \int_0^{\frac{\pi}{2}} \frac{\cos^5 x + \sin^5 x}{\sin^5 x + \cos^5 x} \, dx$
 $= \int_0^{\frac{\pi}{2}} 1 \, dx$
 $= [x]_0^{\frac{\pi}{2}}$
 $= \left[\frac{\pi}{2} - 0 \right]$
 $= \frac{\pi}{2}$
 $\therefore I = \frac{\pi}{4}$

(d) $I = \int_0^{\frac{\pi}{2}} \frac{dx}{2\sin 2x + \cos x}$
 $= \int_0^{\frac{\pi}{2}} \frac{dx}{4\sin x \cos x + \cos x}$
 $= \int_0^{\frac{\pi}{2}} \frac{dx}{\cos x (4\sin x + 1)}$

let $t = \tan \frac{x}{2}$, $x \neq \pi$
 $\therefore \cos x = \frac{1-t^2}{1+t^2}$, $\sin x = \frac{2t}{1+t^2}$
 $dx = \frac{2dt}{1+t^2}$ when $x=0$ $t=0$
 $x=\frac{\pi}{2}$ $t=\tan \frac{\pi}{4}$

$\therefore I = \int_0^{\tan \frac{\pi}{4}} \frac{\frac{2dt}{1+t^2}}{\left(\frac{1-t^2}{1+t^2}\right)\left(\frac{4t}{1+t^2} + 1\right)}$
 $= \int_0^{\tan \frac{\pi}{4}} \frac{2(1+t^2) \, dt}{(1-t^2)(t^2 + 4t + 1)}$

By partial fractions:

$\frac{2(1+t^2)}{(1-t)(1+t)(t^2+4t+1)} = \frac{A}{1-t} + \frac{B}{1+t} + \frac{Ct+D}{t^2+4t+1}$
 $\therefore 2+2t^2 = A(1+t)(t^2+4t+1) + B(1-t)(t^2+4t+1) + (Ct+D)(1-t^2)$

let $t = -1$ $\therefore 4 = -12B$ $\therefore B = -\frac{1}{3}$
 $t = 1$ $\therefore 4 = 12A$ $\therefore A = \frac{1}{3}$
 $t = 0$ $\therefore 2 = A+B+D$ $\therefore D = 2$
 $t = 2$ $\therefore 10 = 63A - 21B - 6C - 3D$
 $\therefore C = 2$

$\therefore I = \int_0^{\tan \frac{\pi}{4}} \frac{\frac{1}{3}}{1-t} - \frac{\frac{1}{3}}{1+t} + \frac{2t+2}{t^2+4t+1} \, dt$
 $= \int_0^{\tan \frac{\pi}{4}} \frac{\frac{1}{3}}{1-t} - \frac{\frac{1}{3}}{1+t} + \frac{4(2t+8)-6}{t^2+4t+1} \, dt$
 $= \int_0^{\tan \frac{\pi}{4}} \frac{\frac{1}{3}}{1-t} - \frac{\frac{1}{3}}{1+t} + \frac{4t+8}{t^2+4t+1} - \frac{6}{t^2+4t+1} \, dt$

$= \left[-\frac{1}{3} \ln|1-t| - \frac{1}{3} \ln|1+t| + \ln|t^2+4t+1| - \frac{6}{2\sqrt{5}} \ln \left| \frac{t+4-\sqrt{5}}{t+4+\sqrt{5}} \right| \right]_0^{\tan \frac{\pi}{4}}$
 $= \left[-\frac{1}{3} \ln|1-\tan \frac{\pi}{4}| - \frac{1}{3} \ln|1+\tan \frac{\pi}{4}| + \ln|1+4+1| - \frac{3}{\sqrt{5}} \ln \left| \frac{1+4-\sqrt{5}}{1+4+\sqrt{5}} \right| \right]$
 $= \ln(1+4+1) + \frac{3}{\sqrt{5}} \ln \left| \frac{4-\sqrt{5}}{4+\sqrt{5}} \right| - \frac{1}{3} \ln|1-\tan^2 \frac{\pi}{4}|$

2(a) $I = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{x^3}{\cos x} \, dx$

let $f(x) = x^3$, $g(x) = \cos x$
 Now as $f(x)$ is odd and $g(x)$ is even
 $\therefore \frac{f(x)}{g(x)}$ is an odd function. (odd fn ÷ even fn is odd fn)
 $\therefore I = 0$ (the integration of an odd function about symmetric limits is zero)

(b) $I = \int \sin^3 2x \cos^2 2x \, dx$
 $= \int \sin 2x (1 - \cos^2 2x) \cos^2 2x \, dx$

let $u = \cos 2x$

$\therefore \frac{du}{dx} = -2 \sin 2x$

$$I = -\frac{1}{2} \int (1-u^2) u^2 du$$

$$= -\frac{1}{2} \int u^2 - u^4 du$$

$$= -\frac{1}{2} \left[\frac{\cos^3 2x}{3} - \frac{\cos^5 2x}{5} \right] + c$$

$$(c) I = \int \frac{4x-3}{\sqrt{6+2x-3x^2}} dx$$

$$= \int \frac{-\frac{2}{3}(-6x+2) - \frac{5}{3}}{\sqrt{6+2x-3x^2}} dx$$

$$= -\frac{2}{3} \int \frac{-6x+2}{\sqrt{6+2x-3x^2}} dx - \frac{5}{3} \int \frac{dx}{\sqrt{6+2x-3x^2}}$$

let $u = \sqrt{6+2x-3x^2}$

$$\therefore \frac{du}{dx} = \frac{1}{2}(-6x+2)$$

$$= -\frac{2}{3} \cdot 2 \int \frac{(-6x+2) dx}{\sqrt{6+2x-3x^2}} - \frac{5}{3} \int \frac{dx}{\sqrt{6+2x-3x^2}}$$

$$= -\frac{4}{3} \sqrt{6+2x-3x^2} - \frac{5}{3} \int \frac{dx}{\sqrt{6+2x-3x^2}}$$

$$= -\frac{4}{3} \sqrt{6+2x-3x^2} - \frac{5}{3\sqrt{3}} \sin^{-1} \left(\frac{x-\frac{1}{3}}{\sqrt{4/3}} \right) + c$$

$$= -\frac{4}{3} \sqrt{6+2x-3x^2} - \frac{5}{3\sqrt{3}} \sin^{-1} \left(\frac{3x-1}{\sqrt{4}} \right) + c$$

$$(d) I_n = \int_0^{\pi/2} \cos^n x \sin^2 x dx, n \geq 0$$

$$= \int_0^{\pi/2} \cos^n x (1 - \cos^2 x) dx$$

$$= \int_0^{\pi/2} \cos^n x - \cos^{n+2} x dx$$

$$= U_n - U_{n+2}$$

(where $U_n = \int_0^{\pi/2} \cos^n x dx$)

Now for $U_n = \int_0^{\pi/2} \cos^n x dx$

$$\therefore U_n = \int_0^{\pi/2} \cos^{n-1} x \cos x dx$$

let $u = \cos^{n-1} x, dv = \cos x dx$

$$\therefore \frac{du}{dx} = (n-1) \cos^{n-2} x \cdot -\sin x \quad v = \sin x$$

$$\therefore U_n = \left[\cos^{n-1} x \sin x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \cos^{n-2} x \sin^2 x dx$$

$$= 0 + (n-1) U_{n-2}$$

$$\therefore U_n = (n-1) U_{n-2}$$

Now as $I_n = U_n - U_{n+2}$

$$\therefore I_n = (n-1) U_{n-2} - (n+1) U_{n+2}$$

$$= (n-1) U_{n-2} - (n+1) I_n$$

$$\therefore I_n (1+n+1) = (n-1) U_{n-2}$$

$$\therefore I_n = \frac{n-1}{n+2} U_{n-2}$$

Now $I_4 = \int_0^{\pi/2} \cos^4 x \sin^2 x dx$

$$= \frac{4-1}{4+2} I_2 \text{ (using the above result)}$$

$$= \frac{3}{2} \cdot \frac{1}{4} \cdot I_0$$

$$= \frac{1}{8} \int_0^{\pi/2} \cos^0 x \sin^2 x dx$$

$$= \frac{1}{8} \int_0^{\pi/2} \frac{1}{2} [1 - \cos 2x] dx$$

$$= \frac{1}{16} \left[x - \frac{\sin 2x}{2} \right]_0^{\pi/2}$$

$$= \frac{1}{16} \left[\left(\frac{\pi}{2} - 0 \right) - 0 \right]$$

$$= \frac{\pi}{32}$$

$$3 (i) (1) z_1 = 1-i \quad |z_1| = \sqrt{2} \quad \arg z_1 = -\frac{\pi}{4} \quad \therefore |z_1 z_2| = 2\sqrt{2} \quad 2$$

$$z_2 = -1+\sqrt{3}i \quad |z_2| = 2 \quad \arg z_2 = \frac{\pi}{3}$$

$$\arg(z_1 z_2) = \frac{\pi}{12} \quad 2$$

$$(ii) z_1 z_2 = \frac{(1-i)(-1+\sqrt{3}i)}{(\sqrt{3}-1) + i(\sqrt{3}+1)}$$

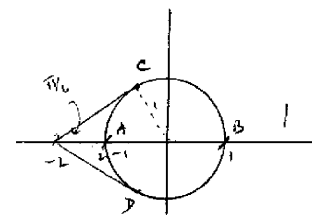
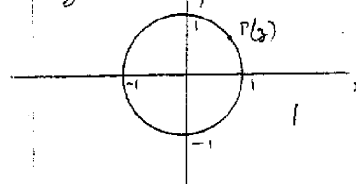
$$\therefore (\sqrt{3}-1) + i(\sqrt{3}+1) = 2\sqrt{2} \left(\cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

Equating real parts

$$\therefore \sqrt{3}-1 = 2\sqrt{2} \cos \frac{\pi}{12}$$

$$\therefore \cos \frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} \quad 2$$

(e) If $|z|=1$
z lies on unit circle



A gives min. value of $|z+2|$ $1 \leq |z+2| \leq 3$
B gives max. value of $|z+2|$
C gives max. value of $\arg(z+2)$ $-\frac{\pi}{2} \leq \arg(z+2) \leq \frac{\pi}{2}$
D gives min. value of $\arg(z+2)$

$$(c) (i) \text{ If } z = x+iy \quad x+iy + \frac{1}{x+iy} = x+iy + \frac{x-iy}{x^2+y^2} = \frac{x(x^2+y^2+1)+iy(x^2+y^2-1)}{x^2+y^2}$$

$$\therefore \text{ If } z + \frac{1}{z} \text{ real} \quad y(x^2+y^2-1) = 0$$

$$\therefore \frac{y}{(x^2+y^2)} = 0 \quad \text{or} \quad \frac{x^2+y^2}{x^2+y^2} = 1 \quad 3$$

(real axis) (unit circle, excluding origin)

$$(ii) \text{ If } y=0 \quad z + \frac{1}{z} = \frac{x(x^2+1)}{x^2} = \frac{x^3+1}{x} \quad \text{and } |z + \frac{1}{z} - 2| = \frac{x^3+1-2x^2}{x^2} = \frac{(x^2-1)^2}{x^2} \geq 0$$

$$\text{If } x^2+y^2=1 \quad |z + \frac{1}{z}| = \frac{|x^3+1|}{x^2} \leq 2 \quad \text{since } |x| \leq 1 \quad \therefore |z + \frac{1}{z}| \geq 2$$

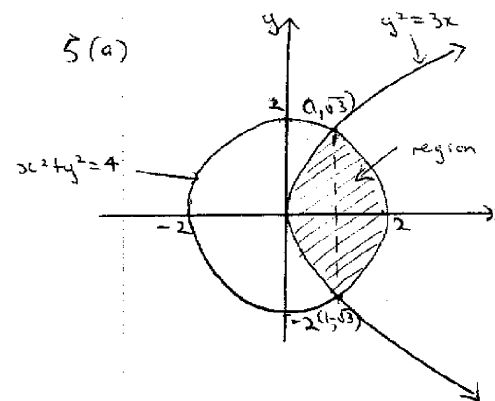
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(a) $P(x) = x^5 + 2x^2 + ax + b$
 $P'(x) = 5x^4 + 4x + a$
 $P'(-1) = P'(-1) = 0$
 $1 - a + b = 0$
 $1 + a = 0$
 $\therefore a = -1, b = -2$

(b) One root = $1 - 2i$
Sum of roots = $-1 - i$
 $\therefore$ Other root = $-1 - i - 1 + 2i$
 $= -2 + i$
And product of roots
 $k = (1 - 2i)(-2 + i)$
 $= -2 + 4i + i + 2 = 5i$

(c) $x^3 + 3px + q = 0$
 $\therefore \alpha + \beta + \gamma = 0$ $\alpha + \beta + \gamma + 2 = 3p$ $2\beta\gamma = -q$
 $\frac{\alpha}{\gamma} + \frac{\beta}{\gamma} + \frac{\gamma}{\gamma} = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma}$
 $= \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma} = -\frac{9p^2}{q}$
 $\frac{\alpha}{\gamma} + \frac{\beta}{\gamma} + \frac{\gamma}{\gamma} = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} = \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma} = -\frac{9p^2}{q}$
 $\frac{\alpha}{\gamma} + \frac{\beta}{\gamma} + \frac{\gamma}{\gamma} = \frac{\alpha^2 + \beta^2 + \gamma^2}{\alpha\beta\gamma} = \frac{(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)}{\alpha\beta\gamma} = -\frac{9p^2}{q}$
 $\therefore$ Monic equation is $x^3 + \frac{9p^2}{q}x^2 - 6px + q = 0$

If $\gamma = \alpha\beta$, one root of this equation is 1
 $\therefore 1 + \frac{9p^2}{q} - 6p + q = 0$
 $\therefore q + 9p^2 - 6pq + q^2 = 0$
 $\therefore (3p - q)^2 + q = 0$



For points of intersection:

$y^2 = 3x$ — (1)

$x^2 + y^2 = 4$ — (2)

sub $y^2 = 3x$ into (2)

$x^2 + 3x - 4 = 0$

$(x + 4)(x - 1) = 0$

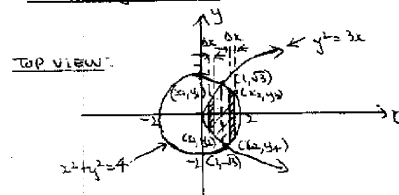
$\therefore x = -4$ or 1

when $x = -4$ y does not exist

$x = 1$ $y = \pm\sqrt{3}$

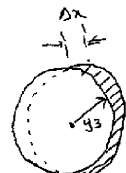
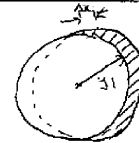
$\therefore$ pts of intersection are $(1, \sqrt{3})$ and $(1, -\sqrt{3})$

(i) by circular discs



Take 2 slices of thickness Δx through the region as shown.

SIDE VIEWS:



Now areas of the cross-sectional slices are:

$A_1(x) = \pi y_1^2$
 $= \pi(3x)$

$A_2(x) = \pi y_2^2$
 $= \pi(4 - x^2)$

Now volume, ΔV , of each slice

$= A(x) \Delta x$
 $\therefore$ Total volume = $\lim_{\Delta x \rightarrow 0} \left[\sum_{x=0}^1 A_1(x) \Delta x + \sum_{x=1}^2 A_2(x) \Delta x \right]$

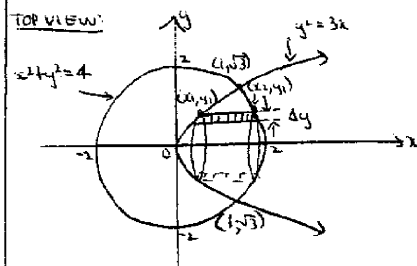
$= \pi \int_0^1 3x dx + \pi \int_1^2 (4 - x^2) dx$

$= \pi \left[\frac{3x^2}{2} \right]_0^1 + \pi \left[4x - \frac{x^3}{3} \right]_1^2$

$= \pi \left[\frac{3}{2} - 0 \right] + \pi \left[\left(8 - \frac{8}{3} \right) - \left(4 - \frac{1}{3} \right) \right]$

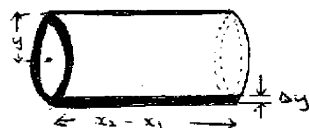
$= \frac{19\pi}{6} \text{ units}^3$

(ii) by cylindrical shells.



Take a slice of thickness Δy perpendicular to the y -axis

Side View:



When the slice is rotated about the x-axis it generates a thin cylindrical shell of area $2\pi r h$.

Now area of shell, $A(y) = 2\pi y (x_2 - x_1)$
 $= 2\pi y (\sqrt{4-y^2} - \frac{y}{3})$

Now volume, ΔV , of each shell $= A(y) \Delta y$
 Total volume $= \lim_{\Delta y \rightarrow 0} \sum_{y=0}^{\sqrt{3}} 2\pi y (\sqrt{4-y^2} - \frac{y}{3}) \Delta y$

$$= 2\pi \int_0^{\sqrt{3}} y (\sqrt{4-y^2} - \frac{y}{3}) dy$$

$$= 2\pi \int_0^{\sqrt{3}} y (4-y^2)^{\frac{1}{2}} dy$$

$$- 2\pi \int_0^{\sqrt{3}} \frac{y^3}{3} dy$$

let $u = (4-y^2)^{\frac{3}{2}}$
 $\frac{du}{dy} = \frac{3}{2}(4-y^2)^{\frac{1}{2}} \cdot -2y$
 $= -3y(4-y^2)^{\frac{1}{2}}$

Total volume $= -\frac{2\pi}{3} \int_0^{\sqrt{3}} -3y(4-y^2)^{\frac{1}{2}} dy$
 $= -\frac{2\pi}{3} \int_0^{\sqrt{3}} y^3 dy$

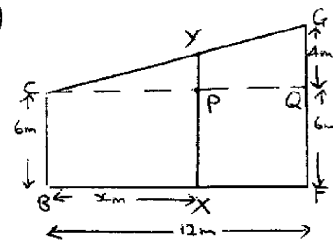
$$= -\frac{2\pi}{3} \left[(4-y^2)^{\frac{3}{2}} \right]_0^{\sqrt{3}} - \frac{2\pi}{3} \left[\frac{y^4}{4} \right]_0^{\sqrt{3}}$$

$$= -\frac{2\pi}{3} [1-8] - \frac{2\pi}{3} \left[\frac{9}{4} - 0 \right]$$

$$= \frac{14\pi}{3} - \frac{3\pi}{2}$$

$$= \frac{19\pi}{6} \text{ units}^3$$

(b)



As $BC \parallel XY \parallel FG$
 $\angle CPY = \angle CQG$ (corr. $\angle$ s bounded by $\parallel$ lines are equal)

similarly $\angle CYP = \angle CQX$

$\angle C$ is common.

$\therefore \triangle CYP \sim \triangle CQX$ (As we have two angles equal)

$\therefore \frac{PY}{4} = \frac{x}{12}$ (corr. sides of similar Δ s are in the same ratio)

$$\therefore PY = \frac{x}{3}$$

$$\therefore XY = 6 + \frac{x}{3}$$

$\Rightarrow$ at a distance x m from the base AB the area of the cross-section is $(6 + \frac{x}{3})^2 \text{ m}^2$.

Now take slices of thickness Δx .

$\therefore$ volume, ΔV , of each slice $= A(x) \Delta x$

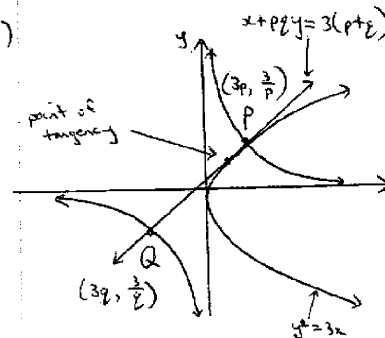
$$\therefore \text{Total volume} = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{12} A(x) \Delta x$$

$$= \int_0^{12} (6 + \frac{x}{3})^2 dx$$

$$= \int_0^{12} 36 + 4x + \frac{x^2}{9} dx$$

$$= \left[36x + 2x^2 + \frac{x^3}{27} \right]_0^{12}$$

6 (a)



(i) $N = \left(\frac{3p+3q}{2}, \frac{\frac{3p}{p} + \frac{3q}{q}}{2} \right)$
 $= \left(\frac{3(pt+q)}{2}, \frac{3(pt+q)}{2pt} \right)$

(ii) $y^2 = 3x$

$$\therefore 2y \frac{dy}{dx} = 3$$

$$\therefore \frac{dy}{dx} = \frac{3}{2y}$$

Let parametric eqns of $y^2 = 3x$

$$\text{be } y = t, x = \frac{t^2}{3}$$

At point of tangency $\frac{dy}{dx} = \frac{3}{2t}$

Eqn of tangent is:

$$y - t = \frac{3}{2t} \left(x - \frac{t^2}{3} \right)$$

$$\therefore 2ty - 2t^2 = 3x - t^2$$

$$\therefore 3x - 2ty = -t^2$$

$$\therefore -\frac{2t}{3}y + x = -\frac{t^2}{3}$$

But PQ is this tangent as well.

$$\therefore -\frac{2t}{3} = p, -\frac{t^2}{3} = 3(pt+q)$$

$$\therefore \frac{3(pt+q)}{2} = -\frac{t^2}{6}$$

Now as $N = \left(\frac{3(pt+q)}{2}, \frac{3(pt+q)}{2pt} \right)$

$$\therefore N = \left(-\frac{t^2}{6}, -\frac{t^2}{6} \cdot \frac{-3}{2t} \right)$$

$$= \left(-\frac{t^2}{6}, \frac{t}{4} \right)$$

$$\therefore x = -\frac{t^2}{6}, y = \frac{t}{4}$$

$$\therefore t = 4y$$

$$\therefore x = -\frac{(4y)^2}{6}$$

$$\therefore 6x = -16y^2$$

$\therefore 3x = -8y^2$ is the locus of N.
 (see over for alternative solution to (a)(ii))

(b) Given: V is constant,

$$\frac{dr}{dt} = 0.05r$$

To Find: $\frac{dh}{dt}$ as a %.

Solution: As $V = \pi r^2 h$

$$\therefore h = \frac{V}{\pi r^2} = \frac{V}{\pi} r^{-2}$$

$$\therefore \frac{dh}{dr} = -\frac{2V}{\pi} r^{-3}$$

$$= -\frac{2V}{\pi r^3}$$

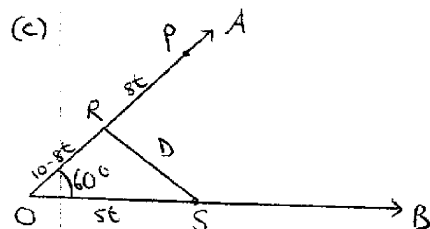
$$\text{Now } \frac{dh}{dt} = \frac{dh}{dr} \frac{dr}{dt}$$

$$\therefore \frac{dh}{dt} = \frac{-2V}{\pi r^3} \times 0.05r$$

$$= \frac{-2\pi r^2 h}{\pi r^3} \times 0.05r$$

$$= -0.1h$$

$\Rightarrow$ height is diminishing at 10% per minute.



After t hours the cyclist has travelled 8t km from P to a point R and the jogger has travelled 5t km from O to a point S. Let $RS = D$.

Now by the cosine rule:

$$D^2 = (10.8t)^2 + (8t)^2 - 2(10.8t)(8t) \cos 60^\circ$$

$$= 100 + 160t + 64t^2 + 25t^2 - 50t + 40t^2$$

$$= 129t^2 - 210t + 100$$

$$\therefore 2D \frac{dD}{dt} = 258t - 210$$

$$\therefore \frac{dD}{dt} = \frac{129t - 105}{D}$$

when $t = 1\frac{1}{2}$ [90 mins = 1.5 hrs]

$$D^2 = 129(1\frac{1}{2})^2 - 210(1\frac{1}{2}) + 100$$

$$= 75.25$$

$$\therefore D = \sqrt{75.25}$$

$$\therefore \frac{dD}{dt} = \frac{129(1.5) - 105}{\sqrt{75.25}}$$

$$= 10.20211039 \dots$$

$$= 10.20 \text{ (2 d.p.)}$$

$\Rightarrow$ distance between the cyclist and jogger is changing at approx 10.20 km/h after 90 mins.

(a)(i) Alternative solution

As $x + py = 3(ptq)$ is a tangent to $y^2 = 3x$ then the quadratic equation formed by substituting $x = \frac{y^2}{3}$ into the equation of this tangent will have a discriminant of zero.

$$\text{i.e. quad. eqn formed is: } \frac{y^2}{3} + py = 3(ptq)$$

$$\therefore y^2 + 3pty - 9(ptq) = 0$$

$$\text{Now } \Delta = (3ptq)^2 - 4(1)(-9(ptq))$$

$$= 9p^2q^2 + 36(ptq)$$

$$\text{But as } \Delta = 0 \therefore p^2q^2 = -4(ptq) \quad \text{--- (1)}$$

Now from the coords of N:

$$y = \frac{3}{4} \left(\frac{ptq}{p^2t^2} \right)$$

$$\therefore y^2 = \frac{9}{16} \left(\frac{(ptq)^2}{p^4t^4} \right)$$

$$= \frac{9}{16} \left(\frac{(ptq)^2}{p^4t^4} \right) \quad (\text{sub. (1)})$$

$$= \frac{-9}{16} (ptq) \quad \text{--- (2)}$$

But also from the coords of N $x = \frac{3(ptq)}{2}$

$$\therefore ptq = \frac{2x}{3} \quad \text{sub into (2)}$$

$$\therefore y^2 = \frac{-9}{16} \left(\frac{2x}{3} \right)$$

$$= -\frac{3x}{8}$$

$$\Rightarrow \text{locus of N is: } 3x = -8y^2$$

$$7. \quad (a) \quad 4x^2 + 9y^2 = 36$$

$$8x + 18 \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{4x}{9y}$$

$$4x^2 - y^2 = 4$$

$$8x - 2y \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = \frac{4x}{y}$$

Solve simultaneously

$$4x^2 + 9y^2 = 36$$

$$4x^2 - y^2 = 4$$

$$10y^2 = 32$$

$$y^2 = \frac{16}{5}$$

$$y = \pm \frac{4}{\sqrt{5}}$$

$$4x^2 = 4 + \frac{16}{5}$$

$$x^2 = \frac{9}{5}$$

$$x = \pm \frac{3}{\sqrt{5}}$$

$$\text{If } x, y \text{ of same sign gradient is } \frac{-4 \cdot \frac{3}{\sqrt{5}}}{\frac{4}{\sqrt{5}}} = -3 \quad \text{and} \quad \frac{4 \cdot \frac{3}{\sqrt{5}}}{\frac{4}{\sqrt{5}}} = 3$$

... opposite sign ...

$\therefore$ Ellipse and Hyperbola intersect at right angle

$$(b) \quad a^2y_1x - b^2x_1y = (a^2 - b^2)x_1y_1$$

For G cut $y = 0$

$$\therefore G \text{ is } \left(\frac{a^2 - b^2}{a^2} x_1, 0 \right) = (e^2 x_1, 0)$$

$$\therefore GS = \left| a(a - ex_1) \right|$$

$$PS = \sqrt{(x_1 - ae)^2 + y_1^2}$$

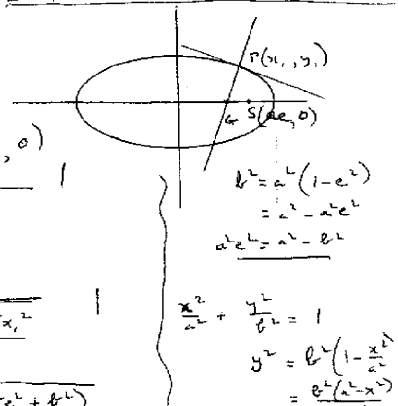
$$= \sqrt{x_1^2 - 2aex_1 + a^2e^2 + \frac{a^2b^2 - b^4x_1^2}{a^2}}$$

$$= \frac{1}{a} \sqrt{x_1^2(a^2 - b^2) - 2a^3ex_1 + a^4(a^2e^2 + b^4)}$$

$$= \frac{1}{a} \sqrt{a^2e^2x_1^2 - 2a^3ex_1 + a^4} = \frac{1}{a} \cdot a \sqrt{e^2x_1^2 - 2ex_1 + a^2}$$

$$= |a - ex_1|$$

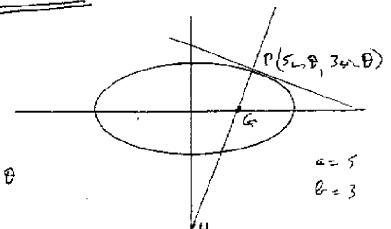
$$\therefore GS = e \times PS$$



(c) Normal is

$$75 \sin \theta \cdot x - 45 \cos \theta \cdot y = 16.15 \sin \theta \cos \theta$$

$$\text{i.e. } 5 \sin \theta \cdot x - 3 \cos \theta \cdot y = 16 \sin \theta \cos \theta$$



$$\therefore G \text{ is } \left(\frac{16 \cos \theta}{5}, 0 \right) \quad | \quad H \text{ is } \left(0, -\frac{16 \sin \theta}{5} \right)$$

$$\therefore \text{Mid point of GH} = \left(\frac{8 \cos \theta}{5}, -\frac{8 \sin \theta}{5} \right)$$

$$\therefore \text{ satisfies } \begin{cases} x = \frac{8 \cos \theta}{5} & \cos \theta = \frac{5x}{8} \\ y = -\frac{8 \sin \theta}{5} & \sin \theta = -\frac{3y}{8} \end{cases}$$

$$\text{ie. } \frac{25x^2}{64} + \frac{9y^2}{64} = 1$$

$$\text{ie. } \frac{x^2}{\frac{64}{25}} + \frac{y^2}{\frac{64}{9}} = 1$$

ie. lies on ellipse

with e given by

$$\frac{64}{25} = \frac{64}{9} (1 - e^2)$$

$$\frac{9}{25} = 1 - e^2$$

$$e^2 = 1 - \frac{9}{25} = \frac{16}{25}$$

$$\therefore e = \frac{4}{5}$$

And original eccentricity given by

$$9 = 25(1 - e^2)$$

$$\frac{9}{25} = 1 - e^2 \quad \therefore e = \frac{4}{5}$$

ie. Same eccentricity

8. (a)

Switch	3. name	Other	No. of ways
0	0	11	0 = 0
0	1	10	${}^2C_0 \times {}^3C_1 \times {}^{10}C_{10} = 3$
0	2	9	${}^2C_0 \times {}^3C_2 \times {}^{10}C_1 = 30$
1	0	10	${}^2C_1 \times {}^3C_0 \times {}^{10}C_{10} = 2$
1	1	9	${}^2C_1 \times {}^3C_1 \times {}^{10}C_1 = 60$
1	2	8	${}^2C_1 \times {}^3C_2 \times {}^{10}C_8 = 270$

$$\text{Total no of ways} = \underline{\underline{365}}$$

(b)

$$\text{Let } \sin^{-1} x = \alpha \quad \cos^{-1} x = \beta$$

$$\therefore x = \sin \alpha \quad x = \cos \beta$$

$$(i) \sin(\sin^{-1} x - \cos^{-1} x) = \sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= x \cdot x - \sqrt{1-x^2} \cdot \sqrt{1-x^2}$$

$$= x^2 - (1 - x^2) = \underline{\underline{2x^2 - 1}} \quad 2$$

(ii)

$$\text{If } \sin^{-1} x - \cos^{-1} x = \sin^{-1}(1-x)$$

$$\sin(\sin^{-1} x - \cos^{-1} x) = 1 - x$$

$$\therefore 2x^2 - 1 = 1 - x$$

$$\therefore 2x^2 + x - 2 = 0$$

$$x = \frac{-1 \pm \sqrt{17}}{4} \quad 2$$

But $\sin^{-1} x$ acute

$$\therefore x + ve$$

$$\therefore x = \frac{-1 + \sqrt{17}}{4}$$

(c)

$$x^2 y^2 - x^2 + y^2 = 0$$

(i)

$$y^2(x^2 + 1) = x^2$$

$$\therefore y^2 = \frac{x^2}{x^2 + 1}$$

$$0 \leq y^2 \leq 1$$

$$0 \leq |y| \leq 1$$

(ii) As $x \rightarrow \pm \infty$ $y^2 \rightarrow 1^- \therefore y \rightarrow 1^-$
or $y \rightarrow -1^+$

$\therefore$ Asymptotes are $y = \pm 1$

$$(iii) \quad x^2 \cdot \frac{\partial^2 z}{\partial x^2} + y^2 \cdot \frac{\partial^2 z}{\partial y^2} - 2x + 2y \frac{\partial z}{\partial x} = 0$$

$$\therefore \frac{dy}{dx} (2x^2y + 2y) = 2x - 2xy^2$$

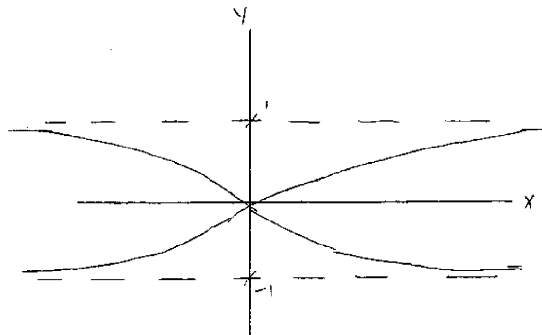
$$\therefore \frac{dy}{dx} = \frac{x(1-y^2)}{y(x^2+1)} \quad 2$$

Ans $y^2(x^2+1) = x^2 \quad \therefore y(21^2+1) = \frac{x^2}{y}$

and $x^2(1-y^2) = y^2 \quad \therefore x(1-y^2) = \frac{y^2}{x}$

$$\frac{dy}{dx} = \frac{y^2}{x} \div \frac{x^2}{y}$$

$$= \frac{5^3}{2 \times 3}$$



(iv)

Note

Even in x and y

∴ Symmetry about both axes

As $x \rightarrow \pm \infty$ $\frac{dy}{dx} \rightarrow 0$